Toward Effective Hyperspectral Image Classification Using Dual-Level Deep Spatial Manifold Representation

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***Abstract*— Hyperspectral image (HSI) contains an abundant spatial structure that can be embedded into feature extraction (FE) or classifier (CL) components for pixelwise classification enhancement. Although some existing works have exploited some simple spatial structures (e.g., local similarity) to enhance either the FE or CL component, few of them consider the latent manifold structure and how to simultaneously embed the manifold structure into both components seamlessly. Thus, their performance is still limited, especially in cases with limited or noisy training samples. To solve both problems with one stone, we present a novel dual-level deep spatial manifold representation (SMR) network for HSI classification, which consists of two kinds of blocks: an SMR-based FE block and an SMR-based CL block. In both blocks, graph convolution is utilized to adaptively model the latent manifold structure lying in each local spatial area. The difference is that the former block condenses the SMR in deep feature space to form the representation for each center pixel, while the later block leverages the SMR to propagate the label information of other pixels within the local area to the center one. To train the network well, we impose an unsupervised information loss on unlabeled samples and a supervised cross-entropy loss on the labeled samples for joint learning, which empowers the network to utilize sufficient samples for SMR learning. Extensive experiments on two benchmark HSI data set demonstrate the efficacy of the proposed method in terms of pixelwise classification, especially in the cases with limited or noisy training samples.**

***Index Terms*— Classification, graph convolution, hyperspectral image (HSI), spatial manifold representation (SMR), small sample.**

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I. Introduction

HYPERSPECTRAL image (HSI) [1], [2] is a kind of digital image, which is obtained by detecting electromagnetic waves reflected by a ground object through a hyperspectral sensor. Different from color or gray-scale images, HSIs usually contain hundreds or even thousands of bands, and each pixel records the spectral curve of the corresponding substance. More importantly, since different substances often show different spectral curves, taking advantage of the spectral information of HSIs is beneficial to accurately distinguishing different substances in the imaging scene. Therefore, with the developing of HSI sensor technology, HSIs have been widely utilized in many civilian and military fields, such as resource exploration [3], [4], environment monitoring [5], [6], and target detection [7]–[9].

In these applications, HSI classification is one of the fundamental tasks, which aims to employ the classifier (CL) trained on the given labeled samples and then assign a class label to unlabeled pixel based on appropriate features. The corresponding methods often consist of a feature extraction (FE) component and a CL component. In the past few decades, diverse HSI classification methods [10]–[12] have been proposed. The early staged approaches are mainly based on traditional pattern recognition methods, such as a K-nearest neighbor (KNN) [13] and a support vector machine (SVM) [14]. While being simple, their classification performance is limited due to their shallow structure. Recently, inspired by the great success of deep neural networks in the computer vision domain, some methods commence at introducing deep neural networks into HSI classification [15]–[17]. Profiting from the deep structure and the flexible fitting capacity, these methods have pushed the HSI classification performance to a higher level.

In both stages, many works [18]–[20] prove that HSIs contain abundant spatial structures, and these structures can be embedded into FE or CL components for pixelwise

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classification enhancement. The reason is intuitive. Due to the spectral similarity and spatial aggregation of homogeneous substances, the spatial structures between different pixels can be well propagated to the class label space and, thus, can be utilized to refine the estimated label of each pixel for performance enhancement. For example, pixels with similar spectral curves in a local spatial domain often have the same class label with high probability. Though these existing works have exploited some local similarities, they focus only on the FE or CL component. Few of them consider the latent manifold structure that is more powerful and complicated [21], [22], as well as how to simultaneously embed the manifold structure into both the FE and CL components seamlessly. Thus, when being confronted with small or noisy labeled samples, their performance is still limited.

To mitigate these limitations, in this study, we present a dual-level deep manifold representation network-based method for HSI classification. First, we build a spatial manifold representation (SMR)-based FE block that employs a graph convolution module to exploit the latent spatial manifold structure in the deep feature domain for each pixel. By doing this, the spatial manifold structure among pixels can be appropriately embedded into a robust feature representation for each pixel. Given these feature representations, we then design two kinds of CLs. One CL is the traditional CL block that predicts the class label. Another one is the SMR-based CL block that employs the graph convolution module to propagate the label information of pixels within a local spatial area to the center one as the final output according to the latent spatial manifold structure. To well-fitting train these three kinds of blocks with sufficient training samples, we propose a semisupervised learning framework that imposes an unsupervised information loss on unlabeled samples and a supervised cross-entropy loss on the labeled samples for joint learning. Extensive experiments on two benchmark HSI data set demonstrate the efficacy of the proposed method in terms of pixelwise classification, especially in the cases with limited or noisy training samples.

In summary, our contribution mainly comes from three aspects as follows.

1) In this article, we present a novel HSI classification approach that can seamlessly embed the complicated spatial manifold structure into both the FE and pixelwise classifications for enhancement. To the best of our knowledge, this is the first attempt to do this in the HSI classification domain.

2) We present a novel semisupervised learning scheme that integrates the conventional cross-entropy loss for labeled samples with an unsupervised information loss for unlabeled ones.

3) We demonstrate state-of-the-art performance on two benchmark HSI classification data sets.

The remainder of this article is organized as follows. In Section II, we introduce the related work. Section III gives the details of the proposed method. Experimental results and analysis are reported in Section IV. Finally, this article is concluded in Section V.

II. Related Work

In this section, we will briefly review the related spectral-based methods and spatial-and-spectral-based methods.

A. HSI Classification Methods

HSI classification methods can be roughly divided into two categories: the conventional pattern recognition-based method and the recent deep learning methods. In the former category, KNN [13] and SVM [14] have been widely used due to their simplicity in both theory and practice. Moreover, the sparse representation and low-rank representation techniques have also been integrated into the HSI classification for performance enhancement [23]. While running efficiently, these conventional methods show limited expressive capacity in characterizing the spectral information for different objects or land-cover categories due to their shallow structure.

To address this problem, the recent deep learning-based methods turned to introduce deep neural networks that show great expressive capacity in HSI classification. In these methods, the stack autoencoder (SAE) [24]–[26] and the convolutional neural network (CNN) [27] are two representative deep learning-based methods. In the SAE [24], the deep neural network is first pretrained in an unsupervised way, and then, a logistic regression (LR)-based CL [28] is added onto the top layer of the network to supervisedly fine-tune the entire network. In the CNN [27], it utilizes a 1-D CNN to extract the feature representation for each pixel and classifies the extracted feature with a fully connected layer. In contrast to the SAE, CNN trains the whole network in a supervised manner and shows more powerful classification capability since it integrates an FE and a CL jointly into one learning framework. Although these deep learning-based methods outperform the conventional pattern recognition-based methods obviously, their performance can be further improved via embedding the latent spatial structure among different pixels into FE or the CL.

B. HSI Classification Enhancement With Spatial Structure

To enhance the performance of HSI classification, various methods have been proposed to embed the spatial structure among different pixels into the FE module or the CL module in the HSI classification. As SVM-based approaches, the spatiospectral Laplacian support vector machine (SS-LapSVM) [18] mines the spatial similarity among different pixels in a local spatial window and then constructs a regularizer to constrain the training process of the conventional SVM-CL. Due to the spatial structure-based regularization, it performs more robust than the conventional SVM-CL, as well as SGSVMSRF [29], which combines the spectral gradient, SVM, and multiscale spatial features fusion with random forest CL into a cascaded framework and obtains higher classification accuracy. In addition, morphological profiles-based methods [30]–[32] have also been proposed to effectively combine spatial and spectral information for HSI classification. For example, in [32], the principal component analysis (PCA) [33] is first used to extract the

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principal components of the spectrum for each pixel as a feature; then, the extended multiattribute morphological profiles (EMAPs) are used to model the spatial information; and, finally, the multinomial logistic regression is utilized as the CL. Furthermore, least-squares regression (LSR)-based CLs are effective in classification tasks. For example, discriminative marginalized least-squares regression (DMLSR) [34] learns a discriminative projection matrix with the constraint of data reconstruction and intraclass compactness graph to avoid overfitting and obtain a near-optimal margin.

Spatial structure is also introduced into the deep learning models. For example, the 2D-CNN [35] extracted local spatial features of each pixel, and the 3D-CNN [36] simultaneously learned the spatial and spectral features of HSI. On this basis, the residual network [37], the dense convolutional network [38], and the capsule network [39] have also been applied to HSI classification tasks. Since the deep learning methods have strong nonlinear representation ability, these methods can effectively improve the classification performance. However, the convolution kernels with a fixed shape, size, and weights are not adaptive to all the regions in an HSI.

The aforementioned spatial structure-based works exploited the local similarity to enhance either the FE or CL component; few of them embedded the manifold structure in both components at the same time. Therefore, in this article, we study the spatial manifold structure in both two stages, by which we can enhance FE and the pixelwise classification.

C. Graph Convolution Network

The graph convolutional network (GCN) [40] is a kind of newly proposed neural network architecture that can perform convolution operation on the graph, which is defined on the spatial domain of a given image through the eigenvector decomposition of the graph Laplacian matrix. Profiting from the flexibility of the graph in terms of representing the latent manifold structure, the GCN can encode complicated relationship among different pixels or local patches into feature representation or inference. Recently, instead of designing graph structure, a nonlocal graph neural network [41] is proposed as a special case of GCN, which computes the response at a position as a weighted sum of the features at all positions. By capturing the long-range dependencies, the computational efficiency can be improved. Thus, a few recent signs of progress of the graph structure try to avoid the explicit graph-based regularization in the loss function via training on a supervised loss for all nodes with labels. In order to further exploit the spatial structures, spectral–spatial GCN (S^2GCN) [42] produces feature by considering both the feature values of adjacency nodes in the graph and the feature values of neighbor pixels in the spatial domain. Moreover, instead of working on a fixed graph, the multiscale dynamic graph convolutional network (MDGCN) [43] flexibly exploit the spatial information based on the irregular image regions (i.e., superpixels). MDGCN adopts the simple linear iterative clustering (SLIC) [44] to segment the entire HSI into a small number of compact superpixels, and each superpixel represents a homogeneous image region with spectral–spatial similarity. Here, the feature of each node is the average spectral signatures of the pixels involved in the corresponding superpixel. Although the GCN-based methods take advantage of the spatial structure information, in these methods, the GCN models are mainly employed in the feature domain. However, the proposed method utilizes the GCN to enhance the FE and CL simultaneously. In addition, most of the GCN-based methods are supervised. To train the network well, we present a semisupervised learning scheme, which empowers the network to use sufficient samples especially in the small samples or the noisy training samples cases.

In practical applications, manual labeling of samples is time-consuming and costly, resulting in very few labeled samples that can be obtained. Furthermore, affected by imaging equipment and the environment, noise is inevitable in the HSI. The small sample size and noise raise challenge deep learning and GCN-based methods, as they will be prone to fall in the overfitting when the model goes deeper, which will seriously affect the generalization ability and classification accuracy of the models. Hence, introducing the unlabeled samples to semisupervisedly train the model is an effective way to improve the classification performance.

III. Proposed Method

This section introduces the proposed dual-level deep SMR network-based hyperspectral imagery classification framework in detail. As shown in Fig. 1, the proposed architecture mainly consists of five blocks, including three spatial manifold representation-based feature extraction (SMR-FE) blocks, one CL block, and one spatial manifold representation-based classifier (SMR-CL) block. First, the SMR-FE block is used to extract the deep feature. Then, the CL block is employed to classify the feature to the class label. Finally, we utilize the SMR-CL block to enhance the classification results. In order to train these five blocks, we build a semisupervised learning framework that integrates the supervised loss on labeled samples with the unsupervised loss on unlabeled samples.

A. Notations

Before introducing the specific blocks, we first introduce the notations. A 3-D HSI cube is often denoted as $\mathcal{X} \in \mathbb{R}^{n_r \times n_c \times n_b}$, which contains n_r rows, n_c columns, and n_b spectral bands. For convenience, we rearrange the 3-D HSI to a 2-D matrix $\mathbf{X} \in \mathbb{R}^{n_p \times n_b}$, where each row stacks a vectorized 2D spatial band image with $n_p = n_r \times n_c$ pixels, and each column denotes one spectrum. $\mathbf{Y} \in \mathbb{R}^{n_p \times L}$ is the corresponding class label for $\mathbf{X}$, in which L is the class number. We randomly split $\mathbf{X}$ and $\mathbf{Y}$ into training and test sets as $\mathbf{X}_{\text{train}} \in \mathbb{R}^{n_{\text{train}} \times n_b}$, $\mathbf{Y}_{\text{train}} \in \mathbb{R}^{n_{\text{train}} \times L}$, $\mathbf{X}_{\text{test}} \in \mathbb{R}^{n_{\text{test}} \times n_b}$, and $\mathbf{Y}_{\text{test}} \in \mathbb{R}^{n_{\text{test}} \times L}$, where n_{train} and n_{test} are the number of training and test samples, respectively.

B. SMR-FE Block

The SMR-FE block is comprised of the convolution module and the GCN module. The convolution module is first utilized to extract the deep features; then, the GCN module is used to aggregate the above output and obtain the more robust and representative features.

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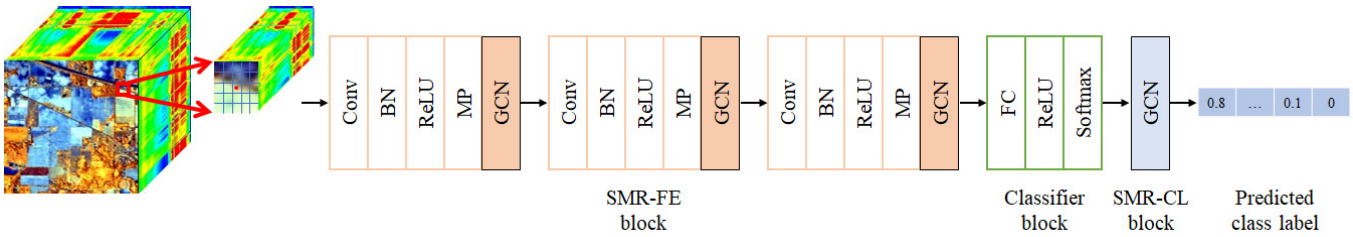


Fig. 1. Proposed architecture. There are five blocks, including three spatial manifold representation-based feature extraction (SMR-FE) blocks, one CL block, and one spatial manifold representation-based classifier (SMR-CL) block.

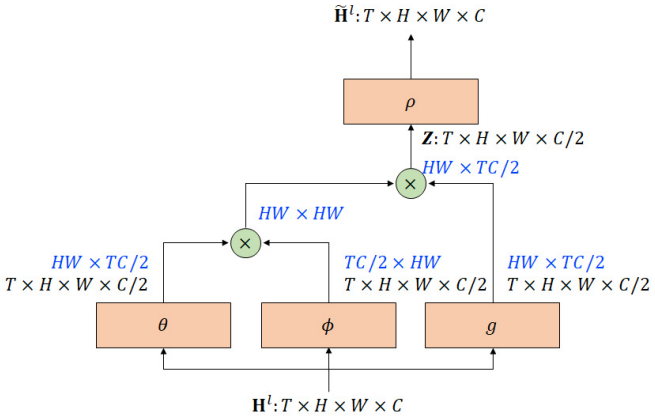


Fig. 2. GCN module of the SMR-FE block. $T \times H \times W \times C$ is the shape of feature maps, T, H, and W represent the spectral and spatial dimensions, and C represents the channels. θ, ϕ, and g represent 3-D convolution operations, and ρ represents 3-D convolution and batch normalization operation.

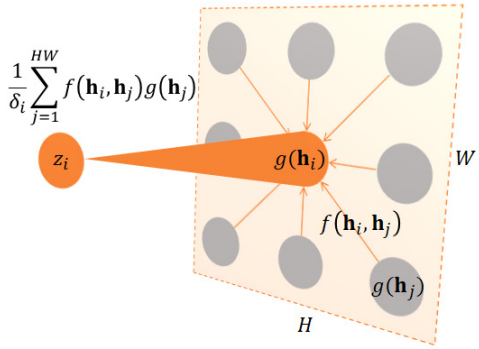


Fig. 3. Locally neighboring pixels used in the GCN module of SMR-FE block. H and W denotes the spatial dimensions. The orange circle ($\mathbf{z}_i$) denotes the output feature. The gray circles ($g(\mathbf{h}_j)$) denote the embedding features for the inputs ($\mathbf{h}_j \in \mathbf{H}^l$). The orange arrows ($f(\mathbf{h}_i, \mathbf{h}_j)$) denote the similarity relationship between any two embedding features (e.g., $\mathbf{h}_i$ and $\mathbf{h}_j$).

To be specific, each convolution module consists of four consecutive submodules, including convolution submodule (Conv), batch normalization submodule (BN), nonlinear activation function submodule, and max-pooling submodule (MP). In this article, the rectified linear unit (ReLU) is adopted as the nonlinear activation function module. For the input image, by convolution module, we can obtain the output feature

$$\mathbf{H}^l = \mathrm{MP}\big(\mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv}\big(\mathbf{H}^{l-1}\mathbf{W}_c^l\big)\big)\big)\big) \tag{1}$$

$\mathbf{H}^{l-1}$ is the input of the lth SMR-FE block, where $l \in \{1, \ldots, n_l - 2\}$, and $n_l = 5$ is the total number of the blocks. $\mathbf{H}^0$ is the original training data. $\mathbf{H}^l$ is the output feature of the convolution module in the lth SMR-FE block. For each pixel, we take its 7×7 neighborhood as the input to extract the feature. $\mathbf{W}_c^l$ is the weight matrix to be learned.

To further enhance the convolution feature, we utilize the GCN module to learn the spatial structure information in the deep feature space and obtain the amendatory feature $\tilde{\mathbf{H}}^l$. To introduce the concept of the GCN operation, we first give the traditional graph convolution propagation rule as [40]

$$\tilde{\mathbf{H}}^l = \rho\big(\hat{\mathbf{A}}\mathbf{H}^l\big) \tag{2}$$

where $\hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$, $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N$, $\mathbf{A}$ is the adjacency matrix of the undirected graph $\mathcal{G}$, $\mathbf{I}_N$ is the identity matrix, $\tilde{\mathbf{D}}$ is the diagonal degree matrix, $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$, and $\rho(\cdot)$ denotes the activation function.

In the traditional GCN module (2), the adjacency matrix $\mathbf{A}$ is always known or manually set. Inspired by [41], we design the proposed GCN module, as shown in Fig. 2. Instead of designing the graph structure, our GCN module can data-driven learn the adjacency matrix in each local spatial domain during network training. The local spatial area pixels used in the GCN module of the SMR-FE block are illustrated in Fig. 3. Specifically, we first map the input features to the embedding spaces and then compute the correlation matrix in the embedding spaces and, finally, obtain the new aggregation results. Hence, $\hat{\mathbf{A}}\mathbf{H}^l$ of (2) is reformulated as $\mathbf{Z}$

$$\mathbf{Z} = \frac{1}{\delta(\mathbf{H}^l)} f\left(\mathbf{H}^l, \mathbf{H}^l\right) g\left(\mathbf{H}^l\right) \tag{3}$$

$$\mathbf{z}_i = \frac{1}{\delta_i(\mathbf{H}^l)} \sum_{\forall j} f\left(\mathbf{h}_i, \mathbf{h}_j\right) g\left(\mathbf{h}_j\right), \quad \forall i. \tag{4}$$

$\mathbf{h}_i, \mathbf{h}_j \in \mathbf{H}^l$ are the input feature, where i and j are the index of the input feature position in spatial domain. The output feature $\mathbf{h}_i \in \mathbf{Z}$ maintains the same size of input $\mathbf{h}_i$. $g(\cdot)$ transfers the input $\mathbf{h}_j$ to the embedding space, and it is implemented as $1 \times 1 \times 1$ convolution operation, i.e., $g\left(\mathbf{h}_j\right) = \mathbf{W}_g^l \mathbf{h}_j$, where $\mathbf{W}_g^l$ is the weight matrix. $f\left(\mathbf{H}^l, \mathbf{H}^l\right)$ is the local correlation matrix, which represents the similarity relationship between any two features (e.g., $\mathbf{h}_i$ and $\mathbf{h}_j$). The more similar feature relative to $\mathbf{h}_i$ has the larger aggregation weight for $g\left(\mathbf{h}_j\right)$. Finally, the output is normalized by a factor $\delta\left(\mathbf{H}^l\right)$.

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We define two kinds of similarity functions $f(\cdot,\cdot)$, i.e., embedded dot product and embedded Gaussian, which are alternative in the proposed GCN module.

1) Embedded Dot Product: The embedded dot product is a kind of simple extension of dot product function to compute similarity in embedding space. $f(\cdot,\cdot)$ can be defined as

$$f\left(\mathbf{h}_i,\mathbf{h}_j\right)=\theta(\mathbf{h}_i)^T\phi\left(\mathbf{h}_j\right). \tag{5}$$

Here, $\theta(\cdot)$ and $\phi(\cdot)$ transfer the corresponding inputs to the embedding spaces, and these two functions are implemented as $1\times1\times1$ convolution operations, i.e., $\theta(\mathbf{h}_i)=\mathbf{W}_\theta^l\mathbf{h}_i$ and $\phi\left(\mathbf{h}_j\right)=\mathbf{W}_\phi^l\mathbf{h}_j$, where $\mathbf{W}_\theta^l$ and $\mathbf{W}_\phi^l$ are the weight matrix. In this case, we set the normalization factor as $\delta\left(\mathbf{H}^l\right)=N_s$ to simplify the computation, where N_s is the number of features in $\mathbf{H}^l$.

2) Embedded Gaussian: Embedded Gaussian takes the linearly embedded Gaussian kernel as the distance metric. It is defined as follows:

$$f\left(\mathbf{h}_i,\mathbf{h}_j\right)=e^{\theta(\mathbf{h}_i)^T\phi\left(\mathbf{h}_j\right)}. \tag{6}$$

Here, the linear embeddings $\theta(\cdot)$ and $\phi(\cdot)$ are defined as above. When the normalization factor is set as $\delta\left(\mathbf{H}^l\right)=\sum_{\forall j}f\left(\mathbf{h}_i,\mathbf{h}_j\right)$, (3) can be represented as the softmax form

$$\mathbf{Z}=\text{softmax}\left(\mathbf{H}^T\mathbf{W}_\theta^{lT}\mathbf{W}_\phi^l\mathbf{H}\right)g(\mathbf{H}). \tag{7}$$

Thus, we can rewrite (2) as

$$\tilde{\mathbf{H}}^l=\rho(\mathbf{Z}). \tag{8}$$

Here, $\tilde{\mathbf{H}}^l$ is the final output of the lth SMR-FE block. $\rho(\mathbf{z}_i)=\mathbf{W}_\rho^l\mathbf{z}_i$, where $\mathbf{W}_\rho^l$ is the weight matrix. In this article, the activation function $\rho(\cdot)$ is implemented as two cascading submodules: the convolution submodule and the batch normalization submodule.

The proposed GCN module is different from [41] in the following two aspects. On the one hand, in order to adapt to the HSI classification task, the proposed GCN module deletes the residual connection "$+\mathbf{H}^l$." However, to accommodate to the video classification task, [41] inserts the residual connection in the GCN module. On the other hand, [41] exploits similar relationships over the global space. In contrast, the proposed GCN module is utilized to adaptively model the manifold structure lying in each local spatial area.

C. Classifier Block

The CL block contains three cascading submodules: the fully connected submodule (FC), ReLU, and the softmax submodule. The fully connected submodule is equivalent to a linear CL. The ReLU is taken as the nonlinear activation function. The softmax submodule is used to ensure that the output of the classifier meets the probability requirement, i.e., each element in the output is nonnegative, and the summation of all elements in the output equals to one

$$\mathbf{P}=\text{softmax}\left(\text{ReLU}\left(\text{FC}\left(\tilde{\mathbf{H}}^{n_l-2}\right)\right)\right). \tag{9}$$

Here, $\mathbf{P}$ is the predicted class label, where $\mathbf{p}_i\in\mathbb{R}^{1\times L}$.

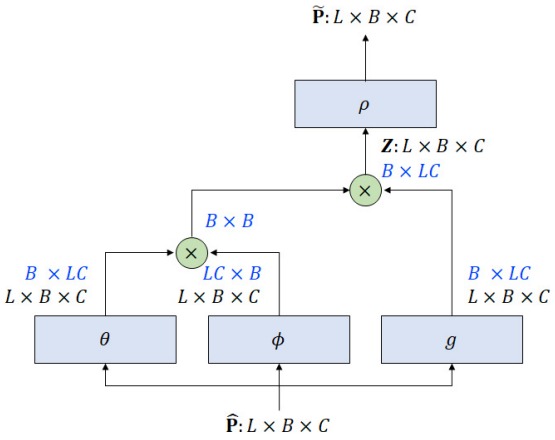


Fig. 4. SMR-CL block. $L\times B\times C$ is the shape of feature maps, L represents the class number, B represents the number of neighborhood samples (i.e., 3×3), and C represents the channels; in this case, $C=1$. θ, ϕ, and g represent 1-D convolution operations, and ρ represents 1-D convolution and batch normalization operation.

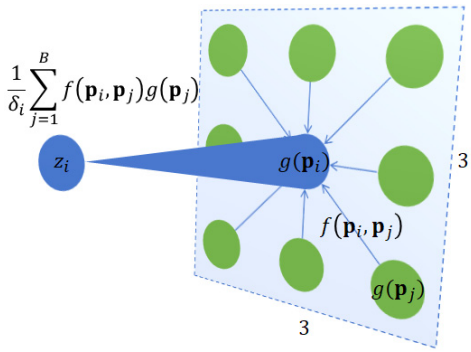


Fig. 5. Locally neighboring pixels used in the SMR-CL block. 3×3 denotes the spatial dimensions. The blue circle ($\mathbf{z}_i$) denotes the output feature. The green circles ($g\left(\mathbf{p}_j\right)$) denote the embedding features for the inputs ($\mathbf{p}_j\in\hat{\mathbf{P}}$). The blue arrows ($f\left(\mathbf{p}_i,\mathbf{p}_j\right)$) denote the similarity relationship between any two embedding features (e.g., $\mathbf{p}_i$ and $\mathbf{p}_j$).

D. SMR-CL Block

Besides the local similarity in the deep feature domain, there are also local similarities in the class label domain. Therefore, based on the spatially neighboring pixels, we build the spatial manifold representation-based classifier (SMR-CL) block to further refine the predicted class label $\mathbf{P}$ and predict the final classification result. Similar to the GCN module of the SMR-FE block, the architecture of the SMR-CL block is shown in Fig. 4, and the local spatial area pixels used in the SMR-CL block are illustrated in Fig. 5.

The input of the SMR-CL block is $\hat{\mathbf{P}}$, which is composed of the predicted class labels of pixel $\mathbf{x}_i\in\mathbf{X}_{\text{train}}$ and its 3×3 neighborhood pixels. For those samples in the 3×3 neighborhood of the center pixel but not in the training set, we put them in the augmented training set $\mathbf{X}_{\text{aug}}\in\mathbb{R}^{n_{\text{aug}}\times n_b}$, where n_{aug} is the number of the augmented unlabeled training samples. It will be trained in the SMR-FE block and the CL block. Thus, by the SMR-CL block, we can obtain the amended class label $\tilde{\mathbf{P}}$

$$\tilde{\mathbf{P}}=\rho(\mathbf{Z}) \tag{10}$$

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$$\mathbf{Z} = \text{softmax}\left(\hat{\mathbf{P}}^T \mathbf{W}_\theta^{n_l T} \mathbf{W}_\phi^{n_l} \hat{\mathbf{P}}\right) g\left(\hat{\mathbf{P}}\right). \tag{11}$$

$g\left(\hat{\mathbf{P}}\right) = \mathbf{W}_g^{n_l} \hat{\mathbf{P}}$, where $\mathbf{W}_g^{n_l}$, $\mathbf{W}_\theta^{n_l}$ and $\mathbf{W}_\phi^{n_l}$ are the weight matrixes.

E. Semisupervised Classification

According to the predicted class label, we construct a supervised loss based on the labeled samples and design an unsupervised loss based on the unlabeled samples. Finally, we optimize the parameters of the SMR-FE block, the CL block, and the SMR-CL block by jointly solving the supervised and unsupervised loss functions.

Specifically, we adopt the cross entropy error over all trained labeled samples as the supervised loss function

$$\$_s = -\sum_{i=0}^{n_{\text{train}}} \mathbf{y}_i \log\left(\mathbf{p}_i\right). \tag{12}$$

Here, $\mathbf{p}_i \in \mathbf{P}$, and $\mathbf{y}_i$ is the corresponding real class label of $\mathbf{x}_i$ $\forall i \in \{1, 2, \ldots, n_{\text{train}}\}$. Minimizing the cross entropy loss could promote the feature learning and increase the probability of assigning the given sample to its ground-truth class.

For the unlabeled samples, the unsupervised loss is calculated by the information entropy

$$\$_{\text{un}} = -\sum_{i=0}^{n_{\text{train}}+n_{\text{aug}}} \mathbf{p}_i \log\left(\mathbf{p}_i\right). \tag{13}$$

Here, $\mathbf{p}_i \in \mathbf{P}$ $\forall i \in \left\{1, 2, \ldots, n_{\text{train}} + n_{\text{aug}}\right\}$.

We integrate the cross-entropy loss with the information entropy loss as the total loss $\$_1$. By optimizing the loss function $\$_1$, we can obtain the optimized parameters of the SMR-FE block and the CL block

$$\$_1 = \$_s + \$_{\text{un}}. \tag{14}$$

Regarding the SMR-CL block, it is trained with the following cross entropy loss:

$$\$_2 = -\sum_{i=0}^{n_{\text{train}}} \mathbf{y}_i \log\left(\tilde{\mathbf{p}}_i\right). \tag{15}$$

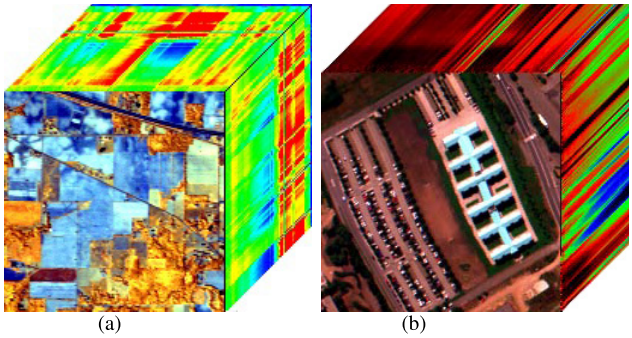
Here, $\tilde{\mathbf{p}}_i \in \tilde{\mathbf{P}}$ $\forall i \in \{1, 2, \ldots, n_{\text{train}}\}$. By optimizing the loss function $\$_2$, we can obtain the optimized parameters of the SMR-CL block.

IV. Experimental Results

A. Data Sets

Two benchmark HSI data sets are adopted for experiments: the Indian Pines data set and the Pavia University data set.

The Indian Pines data set was gathered by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in Northwestern Indiana, USA. There are 145×145 pixels with a spatial resolution of 20 m. The data set captures 220 spectral channels ranging from 0.4 to 2.45 μm, including the visible and infrared spectral regions. After removing the water absorption bands, we reduce the number of spectral channels to 200. To ensure the relative balance of pixel numbers from different classes, as a lot of HSI classification studies [26], [45] adopt on this data set, we discard some classes that only have a few labeled samples and selected nine classes.



(a) (b)

Fig. 6. 3-D cubes of (a) Indian Pines data set and (b) Pavia University data set.

TABLE I

Name and Pixel Numbers of per Adopted Class in the Indian Pines and Pavia University Data Sets

Class	Indian Pines		Pavia University	
	Class Name	Number	Class Name	Number
1	Corn-notill	1428	Asphalt	6431
2	Corn-mintill	830	Meadows	18649
3	Grass-pasture	483	Gravel	2099
4	Hay-windrowed	478	Trees	3064
5	Soybean-notill	972	Metal Sheets	1345
6	Soy-mintill	2455	Bare Soil	5029
7	Soybean-clean	593	Bitumen	1330
8	Woods	1265	Bricks	3682
9	Grass-trees	730	Shadows	947
Total		9234		42776

The Pavia University data set was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in Pavia University, Italy. The image scene has 610×340 pixels with a spatial resolution of 1.3 m. We remove 12 water absorption bands and use the remaining 103 bands for experiments. The spectral range is from 0.43 to 0.86 μm. Approximately, 42 776 labeled pixels with nine classes are from the ground-truth map.

The illustrations of the adopted data sets can be seen from Fig. 6. In addition, the names and the numbers of pixels per adopted class on the above two data sets are listed in Table I.

B. Comparison Methods

To demonstrate the superiority of the proposed method, we compared it with eight state-of-the-art classification methods, including KNN [13], DMLSR [34], SVM [14], SS-LapSVM [18], 1D-CNN [36], 3D-CNN [36], GCN [40], and MDGCN [43]. Among these methods, KNN, DMLSR, SVM, and SS-LapSVM are four kinds of conventional HSI classification methods, in which DMLSR is also a kind of graph-based approach. 1D-CNN and 3D-CNN have three convolution modules and one CL module (same as the CL block of the proposed method). For the proposed methods and 3D-CNN, we take 7×7 neighborhood samples as the

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TABLE II
NETWORK ARCHITECTURE OF THE PROPOSED METHOD

Block	Layer		in-channels	out-channels	kernel-size	stride	padding
SMR-FE1	$Conv$		1	32	(8,1,1)	(3,1,1)	1
	$BN + ReLU$		32	-	-	-	-
	MP		-	-	(2,2,2)	-	-
	GCN	θ	32	16	(1,1,1)	(1,1,1)	0
		ϕ	32	16	(1,1,1)	(1,1,1)	0
		g	32	16	(1,1,1)	(1,1,1)	0
		ρ	16	32	(1,1,1)	(1,1,1)	0
SMR-FE2	$Conv$		32	128	(3,1,1)	(2,1,1)	1
	$BN + ReLU$		128	-	-	-	-
	MP		-	-	(2,2,2)	-	-
	GCN	θ	128	64	(1,1,1)	(1,1,1)	0
		ϕ	128	64	(1,1,1)	(1,1,1)	0
		g	128	64	(1,1,1)	(1,1,1)	0
		ρ	64	128	(1,1,1)	(1,1,1)	0
SMR-FE3	$Conv$		128	256	(3,1,1)	(2,1,1)	1
	$BN + ReLU$		256	-	-	-	-
	MP		-	-	(2,2,2)	-	-
	GCN	θ	256	128	(1,1,1)	(1,1,1)	0
		ϕ	256	128	(1,1,1)	(1,1,1)	0
		g	256	128	(1,1,1)	(1,1,1)	0
		ρ	128	256	(1,1,1)	(1,1,1)	0
Classifier	$FC + ReLU$		256*n [1]	256	-	-	
	$Dropout$		-	-	-	-	p=0.5 [2]
	$softmax$		256	L	-	-	-
SMR-CL	GCN	θ	1	1	1	1	0
		ϕ	1	1	1	1	0
		g	1	1	1	1	0
		ρ	1	1	1	1	0

[1] For Indian Pines dataset n=8, for Pavia University dataset n=4.
[2] p is the dropout probability.

input to extract features. GCN and MDGCN are two kinds of GCN-based methods. The proposed methods are denoted as Ours-EG and Ours-DP, in which EG and DP are the abbreviations of embedded Gaussian and embedded dot products. Table II shows the detailed network architecture of the proposed method. Furthermore, the source codes of DMLSR, GCN, and MDGCN are published by the authors and trained on the same hardware (with GPU) as the proposed methods. We have reimplemented the other methods, and the tuned parameters are adopted for best performance.

C. Evaluation Metric

To quantitatively evaluate the performance of all methods, we adopt three standard measuring criteria: overall accuracy (OA) [14], average accuracy (AA) [14], and KAPPA [46] coefficient. OA denotes the classification accuracy on all testing samples, AA measures the average classwise classification accuracy across all classes, and the KAPPA coefficient calculates the statistic degree of agreement in classification over the expected results and is often normalized to $0 \sim 1$. For each criterion, the higher score denotes the better classification result.

D. Comparison With State-of-the-Art Methods

In this part, we focus on demonstrating the superiority of the proposed method in terms of classification accuracy over the other comparison methods. We conduct HSI classification experiments on the two abovementioned data sets. In the implementation, we randomly select 200 labeled samples per class for training and the remaining samples for test. To reduce the effect of randomness on the classification result, we split training and test data five times to remove the influence of sampling randomness, then we train and test five times on the same data to remove the influence of initial parameter randomness. We report the mean value and standard deviation over 25 experiments for all methods. The experimental results are summarized in Tables III and IV, from which we can obtain the following conclusions.

1) Compared with the traditional pattern recognition methods, i.e., KNN, DMLSR, SVM, and SS-LapSVM, it is clear that the proposed methods are substantially better than these four methods, which demonstrates that the deeply learned feature performs better than the heuristically shallow feature. For example, on the Indian Pines data set, Ours-EG outperforms KNN, DMLSR,

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TABLE III

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE INDIAN PINES DATA SET (200 SAMPLES PER CLASS WERE USED FOR TRAINING)

Class	KNN [13]	DMLSR [34]	SVM [14]	SS-LapSVM [18]	1D-CNN [36]	3D-CNN [36]	GCN [40]	MDGCN [43]	Ours-EG	Ours-DP
1	64.54	83.48	82.32	86.41	79.76	94.96	95.10	93.79	**99.35**	99.27
2	64.71	84.21	87.61	94.69	84.37	99.14	88.58	99.30	**99.89**	99.84
3	92.55	95.76	97.52	97.85	97.20	99.42	95.28	98.37	**99.94**	99.89
4	98.89	99.18	99.16	99.86	98.48	99.81	99.86	99.47	**99.97**	99.97
5	99.39	99.88	99.83	100.00	99.87	99.99	100.00	100.00	**100.00**	100.00
6	76.49	86.70	86.75	90.80	85.83	96.45	95.07	95.73	99.06	**99.16**
7	55.48	72.51	75.47	88.11	69.46	89.78	96.89	96.83	**98.06**	97.69
8	58.23	92.62	92.04	94.27	92.20	99.59	90.16	99.19	99.91	**99.93**
9	93.15	98.26	98.91	99.87	98.89	99.41	97.71	99.85	99.97	**99.97**
AA	78.16 ± 0.93	90.29 ± 1.21	91.07 ± 1.02	94.65 ± 1.02	89.56 ± 3.43	97.62 ± 2.63	95.41 ± 10.81	98.06 ± 0.96	$\mathbf{99.57 \pm 1.81}$	99.52 ± 0.80
OA	72.90 ± 1.54	85.35 ± 3.15	87.37 ± 1.58	92.78 ± 1.28	84.97 ± 4.64	95.89 ± 4.07	95.68 ± 10.88	97.35 ± 1.60	$\mathbf{99.26 \pm 2.75}$	99.15 ± 1.34
Kappa	68.70 ± 1.72	82.65 ± 3.54	85.31 ± 1.80	91.56 ± 1.49	82.56 ± 5.14	95.20 ± 4.73	94.93 ± 12.85	96.82 ± 1.92	$\mathbf{99.13 \pm 3.23}$	99.01 ± 1.57

TABLE IV

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATA SET (200 SAMPLES PER CLASS WERE USED FOR TRAINING)

Class	KNN [13]	DMLSR [34]	SVM [14]	SS-LapSVM [18]	1D-CNN [36]	3D-CNN [36]	GCN [40]	MDGCN [43]	Ours-EG	Ours-DP
1	75.45	83.72	86.16	89.96	86.56	98.30	95.75	96.52	**99.59**	99.53
2	77.74	90.00	92.25	95.89	89.99	99.04	**99.98**	95.81	99.52	99.55
3	77.29	83.83	85.44	92.53	82.53	96.43	87.76	96.67	**99.62**	98.87
4	92.58	94.34	96.65	98.75	96.63	98.54	99.14	95.77	**99.56**	99.31
5	99.23	99.49	99.41	99.54	99.88	100.00	99.96	98.53	99.98	**100.00**
6	76.11	87.03	93.59	98.62	93.14	98.71	97.38	99.76	**99.90**	99.78
7	92.50	94.81	94.34	98.35	93.74	99.63	88.90	99.29	**99.94**	99.82
8	76.85	82.48	86.51	92.87	77.61	95.24	96.28	95.92	98.11	**98.19**
9	99.94	99.95	99.96	99.98	99.96	99.82	**99.99**	95.31	99.96	99.98
AA	85.30 ± 0.86	90.63 ± 1.59	92.70 ± 0.57	96.28 ± 0.58	91.11 ± 4.89	98.41 ± 4.03	96.13 ± 3.26	97.06 ± 0.37	$\mathbf{99.57 \pm 1.83}$	99.45 ± 1.44
OA	79.78 ± 2.04	88.62 ± 2.93	91.41 ± 1.60	95.35 ± 1.41	89.53 ± 5.85	98.45 ± 6.24	97.70 ± 2.27	96.60 ± 1.34	$\mathbf{99.50 \pm 2.83}$	99.44 ± 2.09
Kappa	74.07 ± 2.31	84.92 ± 3.22	88.77 ± 2.00	93.90 ± 1.80	86.37 ± 7.30	97.98 ± 7.99	96.95 ± 3.00	95.46 ± 1.77	$\mathbf{99.34 \pm 3.68}$	99.26 ± 2.72

SVM, and SS-LapSVM by 26.36%, 13.91%, 11.89%, and 6.48% in OA, respectively. Moreover, SS-LapSVM exploits the spatial similarity among local samples and constructs a manifold structure regularizer to constrain the training process of SVM CL. Hence, SS-LapSVM has a better performance than SVM.

2) Compared with deep learning-based methods, namely, 1D-CNN and 3D-CNN, the proposed methods can obtain higher classification results, which demonstrates the effectiveness of the proposed GCN modules. For example, on the Indian Pines data set, compared with 1D-CNN and 3D-CNN, OA of Ours-EG is improved by 14.26% and 3.37%, respectively. As 3D-CNN takes the neighboring samples into the FE stage, it is more robust than 1D-CNN. For example, on the Indian Pines and Pavia University data sets, 3D-CNN outperforms 1D-CNN by 10.92% and 8.92% in OA, respectively.
3) Compared with GCN-based methods, the proposed methods are superior to GCN and MDGCN in all cases, such as on the Indian Pines data set. The improvement of Ours-EG over MDGCN is 1.51% in AA, 1.91% in OA, and 2.31% in Kappa. Since the Indian Pines data set has lower spatial resolution than the Pavia University data set, the classification results on the Pavia University data set are equal or greater than those on the Indian Pines data set in most cases. However, the classification results of MDGCN on the Indian Pines data set are little higher than those on the Pavia University data set. This is because MDGCN is based on superpixel, but the bigger superpixel may introduce more noise (for the Indian Pines data set, the number of samples in each superpixel is about 10. For the Pavia University data set, it is approximately 25.) and cannot increase the classification performance observably.
4) Compared Ours-EG and Ours-DP, the classification accuracy of these two methods remain roughly the same, but the former is little higher. The difference between the embedded dot product version (Ours-DP) and the embedded Gaussian version (Ours-EG) is that the embedded Gaussian version can be represented as the softmax form, which plays the role of an activation function. In brief, the GCN module is not sensitive to these choices, which were also demonstrated in [41].

In order to further illustrate the superiority of the proposed methods, we show the visual classification maps for all methods in Figs. 7 and 8. It can be seen that the proposed methods are closer to the ground truth in most cases, which further validates the effectiveness of the proposed methods. It is worth noting that, as some unlabeled samples were divided into the adjacent superpixel point, the labeled

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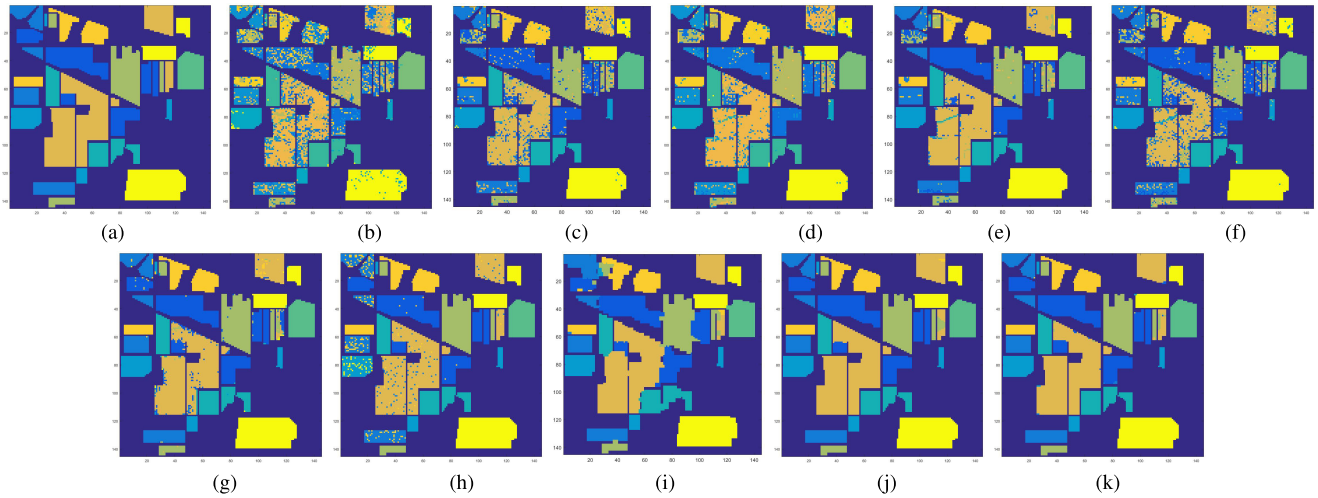


Fig. 7. Classification maps of different methods on the Indian Pines data set (200 samples per class were used for training). (a) Ground truth. (b) KNN. (c) DMLSR. (d) SVM. (e) SS-LapSVM. (f) 1D-CNN. (g) 3D-CNN. (h) GCN. (i) MDGCN. (j) Ours-EG. (k) Ours-DP.

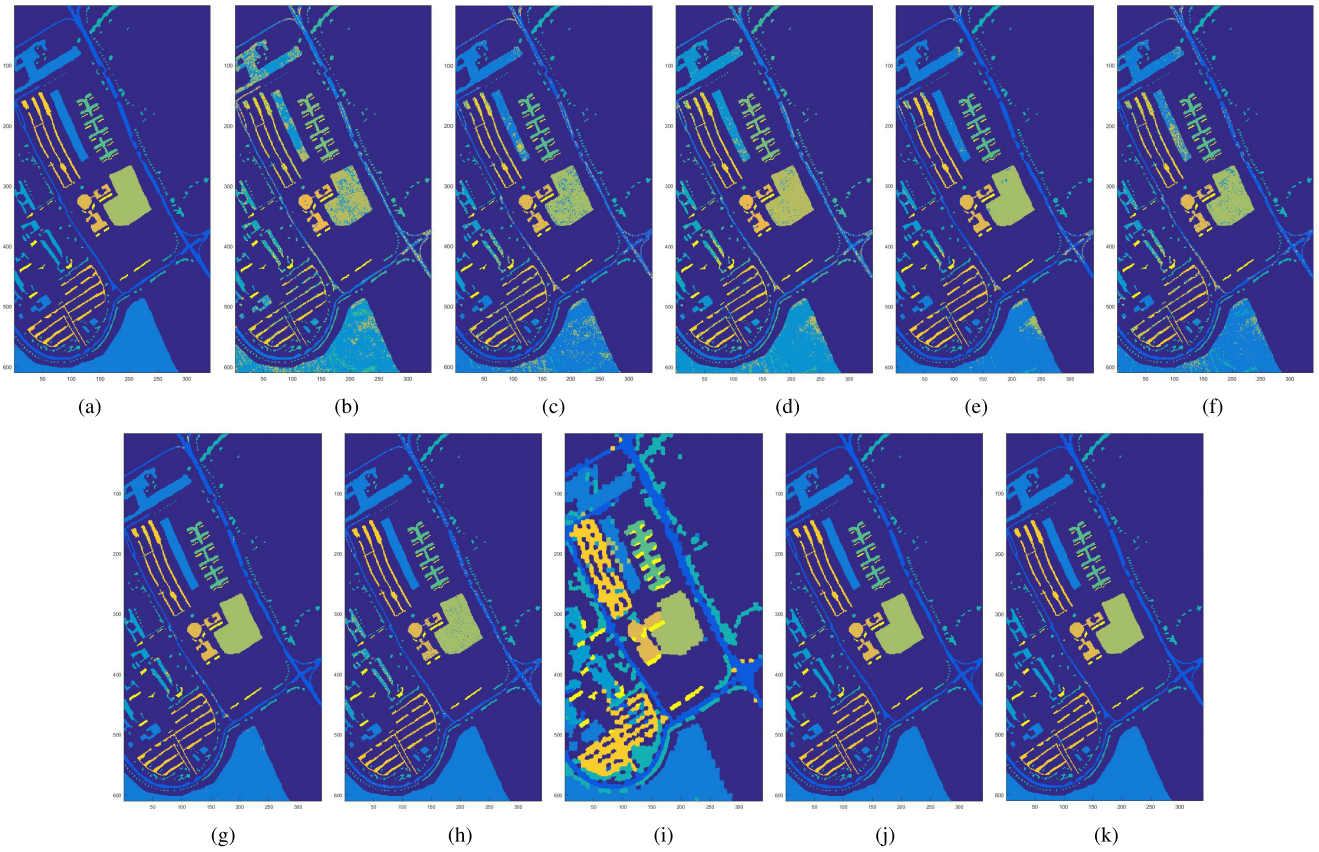


Fig. 8. Classification maps of different methods on the Pavia University data set (200 samples per class were used for training). (a) Ground truth. (b) KNN. (c) DMLSR. (d) SVM. (e) SS-LapSVM. (f) 1D-CNN. (g) 3D-CNN. (h) GCN. (i) MDGCN. (j) Ours-EG. (k) Ours-DP.

zone of MDGCN's classification map is larger than ground truth.

According to the results above, we can conclude that the proposed methods outperform the other eight state-of-the-art competing methods.

E. Robustness to Limited Labeled Samples

To further demonstrate the potential of the proposed methods in dealing with limited labeled samples, we compare the proposed methods with eight state-of-the-art methods (namely, KNN, DMLSR, SVM, SS-LapSVM, 1D-CNN, 3D-CNN, GCN, and MDGCN) on two data sets with different numbers of labeled samples, where the numbers varied from 10 to 200 per class. The classification results are shown in Figs. 9 and 10.

We can see that, as the training set gets smaller, the classification accuracy of each method decreases, as it becomes more difficult to learn discriminative features. The classification

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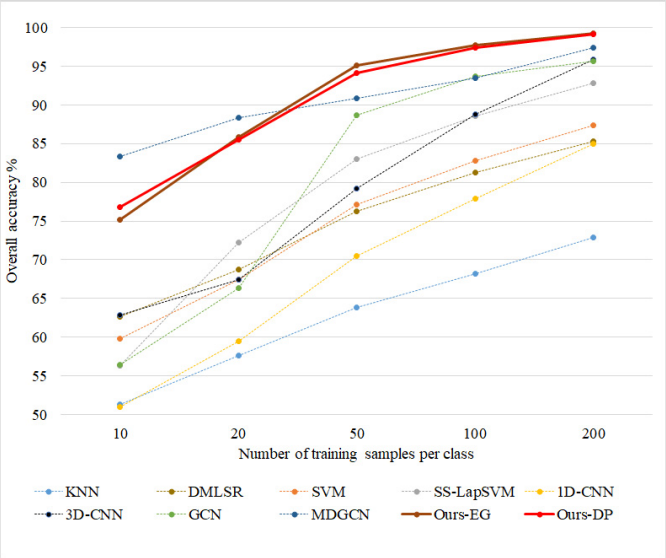


Fig. 9. OA curves for different methods with different numbers of training samples on the Indiana Pines data set.

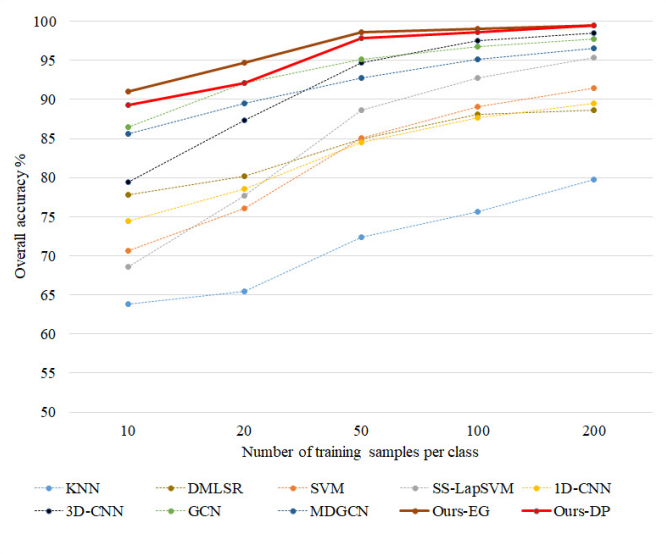


Fig. 10. OA curves for different methods with different numbers of training samples on the Pavia University data set.

results of these methods increase when more samples are introduced for training, which is natural since the CL can be well trained with more training samples. It is worth noting that the proposed methods have superior performance than all competing methods stably given any amount of training samples on the Pavia University data set (see Fig. 10). However, on the Indian Pines data set (see Fig. 9), when the number of training samples per class is less than 20, the classification accuracy of MDGCN is better than that of the proposed methods; the reasons for this are twofold. On the one hand, MDGCN is based on the multiscale dynamic graph to extensively exploit the spatial information, which can discover the hidden spatial context carried by different scales. On the other hand, MDGCN is trained based on superpixels, which actually introduces a lot of samples into the training. For example, there are 9234 labeled samples in the Indian Pines data set, which

TABLE V

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS WITH DIFFERENT LABEL NOISE RATIO ON THE INDIAN PINES DATA SET (50 SAMPLES PER CLASS WERE USED FOR TRAINING)

Ratio		3D-CNN	GCN	Ours-EG	Ours-DP
0%	AA	84.56 ± 3.71	91.71 ± 4.16	$\mathbf{96.83 \pm 2.37}$	96.12 ± 3.52
	OA	79.21 ± 3.77	92.64 ± 5.91	$\mathbf{95.13 \pm 4.54}$	94.10 ± 5.26
	Kappa	75.92 ± 4.23	91.37 ± 6.84	$\mathbf{94.30 \pm 5.27}$	93.10 ± 6.12
10%	AA	83.56 ± 5.09	87.24 ± 8.99	$\mathbf{95.65 \pm 1.50}$	94.87 ± 2.12
	OA	77.88 ± 8.40	88.76 ± 15.83	$\mathbf{93.59 \pm 1.29}$	92.80 ± 3.14
	Kappa	74.37 ± 9.29	86.83 ± 18.09	$\mathbf{92.39 \pm 1.49}$	91.58 ± 3.65
20%	AA	83.73 ± 5.07	86.95 ± 6.31	$\mathbf{94.98 \pm 1.43}$	93.78 ± 1.72
	OA	77.14 ± 7.83	88.28 ± 6.02	$\mathbf{92.02 \pm 2.35}$	91.26 ± 1.37
	Kappa	73.62 ± 8.57	86.23 ± 7.09	$\mathbf{90.69 \pm 2.67}$	89.78 ± 1.61
30%	AA	83.25 ± 2.95	80.24 ± 13.13	$\mathbf{94.29 \pm 4.13}$	94.05 ± 2.33
	OA	77.07 ± 1.75	81.51 ± 21.28	$\mathbf{91.19 \pm 6.09}$	91.07 ± 3.16
	Kappa	73.50 ± 1.91	78.54 ± 24.04	$\mathbf{89.74 \pm 7.04}$	89.57 ± 3.62

were divided into 803 superpixels before training. During a training session, when the number of training samples is 20 per class, with the multiscale dynamic graph and the superpixel mechanism, there are about 1840 samples used for training. Therefore, MDGCN can obtain better classification performance under limited labeled samples case. DMLSR is less sensitive to the number of training samples than other traditional pattern recognition methods. For example, on the Pavia University data set, when the numbers of training samples change from 200 to 10, the OA of DMLSR is decreased by 10.68%; however, the OAs of KNN, SVM, and SS-LapSVM are decreased by 15.92%, 20.74%, and 26.80%, respectively.

In general, we can conclude that the proposed methods are effective for HSI classifications with limited labeled samples in most cases.

F. Robustness to Label Noise

Aiming for validating the robustness of the proposed methods with noisy training samples, we conduct experiments with 3D-CNN and GCN, under different ratios of label noise. Specifically, we randomly select a certain percentage of labeled samples from the training set; then, we assign new labels to these selected samples, where the label noise ratio is set as 0%, 10%, 20%, and 30%, and the new labels are randomly generated within the range $\{1, 2, \ldots, L\}$.

Tables V and VI show the classification results for each method on two data sets. When the label noise ratio increases from 0% to 30%, the classification accuracy decreases for all the methods. However, the proposed methods are robust to the label noise to a certain extent, which can maintain pretty high accuracy. For example, on the Pavia University data set, 3D-CNN drops from 94.21% to 91.04% in AA, GCN drops from 95.27% to 89.04% in AA, and Ours-EG drops from 98.50% to 96.38% in AA. In addition, with the increase in the label noise ratio, the classification accuracy of GCN decreases the most, which is because the label noise has a strong impact on the graph structure. For GCN, the inputs (nodes) are the deep features, and the adjacency matrices (edges) are generated according to the class labels. In contrast, the inputs of the proposed GCN modules are the deep features or class probabilities of the center pixel and its spatially neighboring

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TABLE VI

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS WITH DIFFERENT LABEL NOISE RATIO ON THE PAVIA UNIVERSITY DATA SET (50 SAMPLES PER CLASS WERE USED FOR TRAINING)

Ratio		3D-CNN	GCN	Ours-EG	Ours-DP
0%	AA	94.21 ± 3.46	95.27 ± 1.48	**98.50 ± 2.35**	97.82 ± 2.19
	OA	94.74 ± 3.91	96.87 ± 0.32	**98.62 ± 2.79**	97.86 ± 3.05
	Kappa	93.14 ± 5.01	95.86 ± 0.43	**98.19 ± 3.62**	97.21 ± 3.94
10%	AA	94.11 ± 3.05	91.45 ± 5.16	96.58 ± 1.81	**97.19 ± 3.21**
	OA	94.28 ± 5.31	93.40 ± 3.33	97.30 ± 2.58	**97.38 ± 3.05**
	Kappa	92.55 ± 6.70	91.30 ± 4.37	96.45 ± 3.43	**96.58 ± 3.98**
20%	AA	92.25 ± 8.12	90.60 ± 7.53	96.39 ± 4.25	**96.44 ± 2.64**
	OA	93.18 ± 5.19	92.70 ± 5.64	**97.12 ± 4.22**	96.77 ± 3.33
	Kappa	91.09 ± 6.91	90.37 ± 7.39	**96.20 ± 5.58**	95.77 ± 4.31
30%	AA	91.04 ± 10.38	89.04 ± 4.91	**96.38 ± 4.83**	94.22 ± 6.86
	OA	92.15 ± 9.33	91.05 ± 6.84	**96.89 ± 3.63**	94.33 ± 9.80
	Kappa	89.77 ± 11.89	88.19 ± 8.85	**95.93 ± 4.77**	92.63 ± 12.61

pixels, and the adjacency matrices are computed by similarity relationships of the inputs. Thus, the wrong labels have a lesser impact on the structure of the proposed GCN modules, which results in that the proposed methods can alleviate the impact of label noise.

G. Ablation Study

1) Performance Evaluation With Different GCN Modules: In this part, we conduct the ablation study to validate the effectiveness of the proposed GCN modules in the SMR-FE block and the SMR-CL block. Specifically, we compare SMR-FE with its three variants, including 3D-CNN, Ours-EG1, and Ours-EG2. 3D-CNN is the simplified version of the proposed method, without the GCN module in both SMR-FE block and SMR-CL block. When we delete the GCN module in the SMR-FE block, the proposed method Ours-EG degenerates into Ours-EG1, and when we omit the SMR-CL block, Ours-EG degenerates to Ours-EG2. In the implementation, we randomly select 50 labeled samples per class for training and the remaining samples for testing. The other experimental settings are the same as in Section IV-D.

Table VII reports the classification results of different methods on the Indian Pines data set.

1) Compared Ours-EG2 with 3D-CNN, as can be seen, the GCN module of the SMR-FE block improves the OA up to 11.45%. Because the GCN module of the SME-FE block can contribute to obtain a manifold structure in the deep feature domain and acquire better feature representation.
2) Compared Ours-EG1 with 3D-CNN, or Ours-EG with Ours-EG2, it can be concluded that the SMR-CL block can effectively improve the classification performance of the model, as it integrates the latent manifold structure into CL for enhancement.
3) Compared Ours-EG with 3D-CNN, the classification accuracy of 3D-CNN is lower since Ours-EG considers latent manifold structure in the deep feature domain and the class label domain; both of these advantages benefit the proposed method in improving classification performance.

TABLE VII

CLASSIFICATION ACCURACY (%) OF OURS-EG AND ITS THREE VARIANTS WITH DIFFERENT GCN MODULES ON THE INDIAN PINES DATA SET (50 SAMPLES PER CLASS WERE USED FOR TRAINING)

Class	3D-CNN	Ours-EG1	Ours-EG2	Ours-EG
1	69.20	76.07	88.77	**95.18**
2	73.48	85.13	92.87	**97.68**
3	93.85	96.17	95.90	**98.13**
4	98.82	**99.83**	97.72	99.32
5	99.77	**100.00**	99.87	99.97
6	78.06	83.77	91.12	**94.95**
7	62.32	76.34	81.47	**88.69**
8	86.85	95.23	95.46	**98.66**
9	98.68	**99.63**	96.85	98.91
AA	84.56 ± 3.71	90.24 ± 6.84	93.34 ± 4.80	**96.83 ± 2.37**
OA	79.21 ± 3.77	86.44 ± 9.54	90.66 ± 6.09	**95.13 ± 4.54**
Kappa	75.92 ± 4.23	84.19 ± 10.92	89.10 ± 7.05	**94.30 ± 5.27**

TABLE VIII

CLASSIFICATION ACCURACY (%) OF OURS-EG AND ITS THREE VARIANTS WITH DIFFERENT GCN MODULES ON THE PAVIA UNIVERSITY DATA SET (50 SAMPLES PER CLASS WERE USED FOR TRAINING)

Class	3D-CNN	Ours-EG1	Ours-EG2	Ours-EG
1	91.46	96.52	96.63	**98.48**
2	97.33	98.06	98.01	**99.13**
3	86.79	92.22	92.89	**95.99**
4	95.46	97.33	95.96	**98.89**
5	99.96	**99.98**	99.04	99.92
6	98.60	98.88	96.09	**99.22**
7	95.90	98.20	98.94	**99.67**
8	83.14	89.03	93.74	**95.67**
9	99.30	**99.75**	98.56	99.54
AA	94.21 ± 3.46	96.66 ± 6.02	96.65 ± 11.04	**98.50 ± 2.35**
OA	94.74 ± 3.91	96.89 ± 6.68	96.86 ± 9.62	**98.62 ± 2.79**
Kappa	93.14 ± 5.01	95.94 ± 8.62	95.88 ± 12.99	**98.19 ± 3.62**

A similar conclusion on the Pavia University data set also can be seen in Table VIII.

From the above experiments, we can see that both two kinds of GCN modules in the SMR-FE block and the SMR-CL block can help to address the overfitting problem and improve the classification performance.

2) Performance Evaluation With Different Spatial Sizes of the Inputs in SMR Blocks: We perform experiments to validate the influence of the spatial size of the inputs in two kinds of SMR-based blocks. We fix the spatial size of the input in the SMR-CL block as 3×3 and change the spatial size of the input in the SMR-FE block from 3×3 to 9×9. The classification results of Ours-EG on the Indian Pines data set are summarized in Table IX. We can see that, with the increase in the spatial size, the classification results are increasing first. However, when it surpasses 7×7, there is no significant increase in accuracy. In addition, considering the cost of time and memory, thus, we set the spatial size of the input in the SMR-FE block as 7×7.

The classification results of Ours-EG with different spatial sizes of the inputs in the SMR-CL block on the Indian Pines data set are summarized in Table X. In this part, we set the

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TABLE IX

CLASSIFICATION ACCURACY (%) OF OURS-EG WITH DIFFERENT INPUT SPATIAL SIZES OF THE SMR-FE BLOCK ON THE INDIAN PINES DATA SET (50 SAMPLES PER CLASS WERE USED FOR TRAINING). THE SPACE SIZE OF THE INPUT IN SMR-CL IS 3×3

Class	3×3	5×5	7×7	9×9
1	64.08	82.58	**95.18**	94.66
2	80.31	95.03	97.68	**99.47**
3	92.99	94.57	**98.13**	95.65
4	99.12	**99.51**	99.32	99.26
5	99.33	100.00	99.97	**100.00**
6	80.58	90.20	**94.95**	94.17
7	71.95	73.35	**88.69**	88.01
8	91.23	98.00	98.66	**99.75**
9	98.29	99.11	98.91	**99.65**
AA	86.43 ± 3.22	92.48 ± 3.56	$\mathbf{96.83 \pm 2.37}$	96.74 ± 2.12
OA	81.94 ± 5.89	88.19 ± 4.43	$\mathbf{95.13 \pm 4.54}$	94.97 ± 2.83
Kappa	79.01 ± 6.61	86.28 ± 5.09	$\mathbf{94.30 \pm 5.27}$	94.10 ± 3.32

TABLE X

CLASSIFICATION ACCURACY (%) OF OURS-EG WITH DIFFERENT SPATIAL SIZES OF THE SMR-CL BLOCK ON THE INDIAN PINES DATA SET (50 SAMPLES PER CLASS WERE USED FOR TRAINING). THE SPACE SIZE OF THE INPUT IN SMR-FE BLOCK IS 7×7

Class	3×3	5×5	7×7
1	**95.18**	94.68	93.38
2	97.68	98.55	**99.03**
3	**98.13**	95.80	95.71
4	99.32	99.34	**99.67**
5	99.97	**100.00**	99.96
6	**94.95**	91.14	92.70
7	**88.69**	86.63	83.89
8	98.66	**99.58**	99.10
9	98.91	**99.70**	99.27
AA	$\mathbf{96.83 \pm 2.37}$	96.16 ± 1.38	95.86 ± 5.71
OA	$\mathbf{95.13 \pm 4.54}$	94.22 ± 1.81	93.42 ± 9.87
Kappa	$\mathbf{94.30 \pm 5.27}$	93.24 ± 2.11	92.34 ± 11.38

spatial size of the input in the SMR-FE block as 7×7 and change the spatial size of the input in the SMR-CL block from 3×3 to 7×7. It can be observed that a larger neighborhood scale of the SMR-CL block does not necessarily result in better performance since it will introduce more dissimilar pixels and impact the classification results. Moreover, a larger neighborhood size will result in more parameters, which will demand more samples to train the network to avoid overfitting. Therefore, we set the spatial size of the input in the SMR-CL block as 3×3

3) Effect of the Residual Connection in the GCN Modules: In this part, we experiment to verify the effect of the residual connections in the proposed GCN modules. We use "w/o rc" to denote the proposed method without residual connections on both SMR-FE and SMR-CL blocks, i.e. Ours-EG. "SMR-FE w rc" denotes Ours-EG with residual connection on the SMR-FE block but without residual connection on the SMR-CL block. "SMR-CL w rc" denotes Ours-EG with residual connection on the SMR-CL block but without residual connection on the SMR-FE block. "w rc" denotes Ours-EG with residual connections on both SMR-FE and SMR-CL blocks. The corresponding results on the Indian Pines data set can be seen in Table XI. The results show that adding the residual connections on SMR-FE block, SMR-CL block, or both two blocks will reduce the classification accuracy. Therefore, in the proposed methods, we discard the residual connections on both SMR-FE and SMR-CL blocks.

TABLE XI

CLASSIFICATION ACCURACY (%) OF OURS-EG WITH OR WITHOUT RESIDUAL CONNECTION ON SMR-FE AND SMR-CL BLOCKS ON THE INDIAN PINES DATA SET (20 SAMPLES PER CLASS WERE USED FOR TRAINING)

Class	w/o rc	SMR-FE w rc	SMR-CL w rc	w rc
AA	**90.09**	86.17	86.43	82.88
OA	**85.90**	83.81	84.19	79.64
Kappa	**83.62**	81.03	81.47	76.27

V. CONCLUSION

In this study, we present a dual-level deep SMR network-based HSI classification framework, which obviously improves the classification accuracy, especially when the labeled samples are limited and with label noise. The proposed network consists of two kinds of SMR-based blocks. One block is the SMR-based FE block, which utilizes the graph convolution module to exploit the latent manifold structure among different pixels and embed the structure into a more discriminative feature representation for each pixel. The other one is the SMR-based CL block that aims at integrating the latent manifold structure into CL for enhancement. To appropriately train the proposed network with sufficient samples, we further present a semisupervised learning framework, which separately imposes the conventional cross-entropy loss and an unsupervised information loss on the labeled and unlabeled samples for joint learning. Extensive experimental results on two benchmark data sets demonstrate the superiority of the proposed framework over several state-of-the-art methods.

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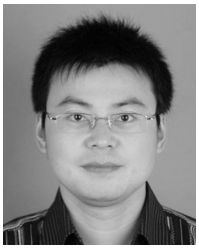
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